Discourse-JEPA:
Latent World Models over Reasoning Steps

Method and first two experimental cycles — fully reproducible from these slides

TextJEPA project

July 9, 2026 (repo: `TextJEPA/`, runs: `runs/disc_*`)

Idea in one slide

- Treat a **reasoning trace** as trajectories of a world model: the *state* s_t is a compressed discourse state ("what has been established"), the *action* a_t is a tiny latent code of an *intent phrase* for the next step.
- Actions state intent only (*"derive purple apples from orange jars times green pens ."*) — never the outcome ("$= 5$"). The predictor $F(s_t, a_t)$ must therefore **compute consequences in latent space**.
- Generation = **search over actions**: roll candidate actions forward with F, score with an energy, execute the best in the environment, re-encode, replan (closed-loop MPC). **No decoder, no token loss anywhere** (JEPA-pure; the symbolic environment renders chosen actions via templates).
- Synthetic reasoning world $\Rightarrow$ exact ground truth for **linear probes** and **objective planning success**.

World: iGSM-style synthetic reasoning (exact generator)

Problem sampler (rejection-sampled until $3 \le |\mathcal{A}_q| \le 9$; `rng = Random(f"{seed}:{index}")`):

- $n \sim \mathcal{U}\{6, \dots, 12\}$ variables; names = distinct (adjective, noun) pairs from 20×20 vocabulary.
- Variable i: with p=0.35 (always for $i < 2$) a **leaf** $x_i = c_i,\ c_i \sim \mathcal{U}\{0..22\}$; else an **internal** $x_i = x_a \circ x_b \bmod 23$, $\circ \in \{+, -, \times\}$, parents a, b sampled from $\{0..i-1\}$.
- Query q = uniform over internal variables; $\mathcal{A}_q$ = ancestors of q (incl. q) = the necessary computation; the rest are **distractors**.

Training traces: solve by repeatedly resolving a feasible variable (parents resolved); with p=0.15 (max 2) pick a feasible *distractor* instead of a necessary one $\Rightarrow$ off-path states for the value head. Fresh problems every epoch (train seed 1, val seed 2; 100k/5k).

A verbatim sample (val problem #3)

Prompt (definitions shuffled per-sample, question last; one sentence = one chunk):

the number of purple apples equals the number of orange jars times the number of green pens .
the number of orange jars is 15 . *the number of green pens is 8 .*
(+ 7 distractor definitions) *how many purple apples are there ?*

Solution trace — step sentence (observed outcome) vs. action intent phrase (no outcome):

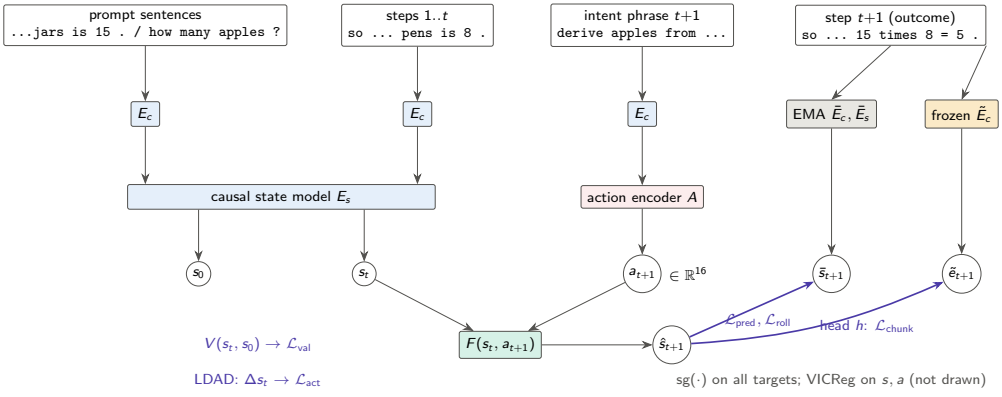
step sentence (encoder sees)	intent phrase (action encoder sees)
so the number of green pens is 8 .	look up the number of green pens .
so the number of orange jars is 15 .	look up the number of orange jars .
so the number of purple apples is 15 times 8 = 5 .	derive purple apples from orange jars times green pens .

Note $15 \times 8 = 120 \equiv 5 \pmod{23}$: outcomes are nontrivial to predict.

Architecture (all sizes exact; 8.8–9.1M params)

- **Chunk encoder** E_c: word-level vocab (441 tokens), embed 256, 2-layer pre-norm transformer (4 heads, FF 1024), masked mean-pool → one 256-d vector per sentence. Shared for prompt, steps, intents.
- **State model** E_s: 4-layer *causal* transformer (8 heads, FF 1024) over [prompt chunks | step chunks] + learned positions + 2 segment embeddings. s_0 = hidden at the question; s_t = hidden at step t.
- **Action encoder** A: MLP 256 → 256 → 16 on the intent-phrase embedding (optional FSQ, unused in main runs).
- **Predictor** F: residual MLP, $\hat{s}_{t+1} = s_t + \mathrm{MLP}([\mathrm{LN}(s_t); a_t])$, 2 hidden layers of 1024 (GELU).
- **Hierarchy**: CLS-transformer over K=3 action codes → 8-d macro action m_t; $F_{\mathrm{hi}}(s_t, m_t)$ predicts $\bar{s}_{t+3}$ (same latent space, HWM-style).
- **LDAD** (Delta-JEPA): MLP on $\Delta s = s_{t+1} - s_t$ → op logits (4) + regression to the EMA intent embedding.
- **Value head** V: MLP on $[\mathrm{LN}([s_t; s_0])]$ → scalar = predicted remaining necessary steps.
- **Teachers**: EMA copies of E_c, E_s (momentum 0.99 → 0.999 linear); **frozen anchor** $\tilde{E}_c$ = copy of E_c at random init, never updated.

Training pipeline



Objectives (exact losses and weights)

Let $d(x, y) = \mathrm{smoothL1}(\mathrm{LN}(x), \mathrm{LN}(y))$ (per-dim mean; LN = parameter-free layer norm), masks over valid steps.

$$\mathcal{L}_{\text{pred}} = d\big(F(s_t, a_t),\ \mathrm{sg}\,\bar{s}_{t+1}\big) \qquad \text{weight } 1.0$$

$$\mathcal{L}_{\text{roll}} = d\big(F^{(t)}(s_0, a_{1:t}),\ \mathrm{sg}\,\bar{s}_t\big) \text{ (open loop, all } t) \qquad 1.0$$

$$\mathcal{L}_{\text{hi}} = d\big(F_{\text{hi}}(s_t, m_t),\ \mathrm{sg}\,\bar{s}_{t+3}\big) \qquad 0.5$$

$$\mathcal{L}_{\text{vic}} = \textstyle\sum_j \max(0, 1 - \mathrm{std}(s^{(j)})) + 0.04\,\|\mathrm{offdiag}(\mathrm{Cov}(s))\|_F^2/D + 0.1\,\mathrm{var}(a) \qquad 1.0$$

$$\mathcal{L}_{\text{act}} = \mathrm{CE}\big(\mathrm{op} \mid \Delta s_t\big) + d\big(g(\Delta s_t),\ \mathrm{sg}\,\bar{E}_c(u_t)\big) \quad \text{(Delta-JEPA)} \qquad 2.0$$

$$\mathcal{L}_{\text{val}} = \mathrm{smoothL1}\big(V(s_t, s_0),\ \#\text{unresolved necessary}\big) \qquad 1.0$$

$$\mathcal{L}_{\text{chunk}} = d\big(h(\hat{s}_{t+1}),\ \mathrm{sg}\,\tilde{E}_c(y_{t+1})\big) + 0.5\,d\big(h(\text{rollout}_t),\ \mathrm{sg}\,\tilde{E}_c(y_t)\big) \qquad 0\ /\ 2.0$$

$$\mathcal{L}_{\text{curv}} = 1 - \cos(v_t,\ v_{t+1}), \quad v_t = s_{t+1} - s_t \text{ (straightening, arXiv:2603.12231)} \qquad 0\ /\ .02\text{–}.1$$

$$\mathcal{L}_{\text{mono}} = \mathbb{1}_{\text{nec}}\,[\Delta d_t^{\text{goal}} + m]_+ + \tfrac{1}{2}(1 - \mathbb{1}_{\text{nec}})\,[-\Delta d_t^{\text{goal}}]_+ \qquad 0\ /\ 1\text{–}2$$

$\mathcal{L}_{\text{chunk}}$: weight 0 in `disc_base`; 2.0 in `disc_chunkpred/combo` (**frozen** anchor $\tilde{E}_c$). Value head input detached except `disc_valgrad/combo`. d_t^{goal} = LN-L1 distance of s_t to the trace-terminal EMA state; margin $m \in \{.02, .05\}$.

Training protocol

- AdamW, lr 3×10^{-4}, weight decay 0.05 (none for dim <2 params), $\beta=(0.9,0.95)$; cosine schedule to floor 0.05, 1000-step warmup; grad clip 5.0.
- Batch 128 problems; 20 epochs $\times$ 781 steps; **fresh data every epoch** (no memorization confound). EMA momentum 0.99 $\rightarrow$ 0.999 linear over training.
- Hardware: one V100-32GB, fp32, $\sim$40 min/run.
- In-training eval each epoch: all losses + collapse diagnostics (per-dim std, RankMe effective rank) + closed-form ridge probes.
- Code: `scripts/train.py +experiment=...`; every run's exact config is stored in its checkpoint.

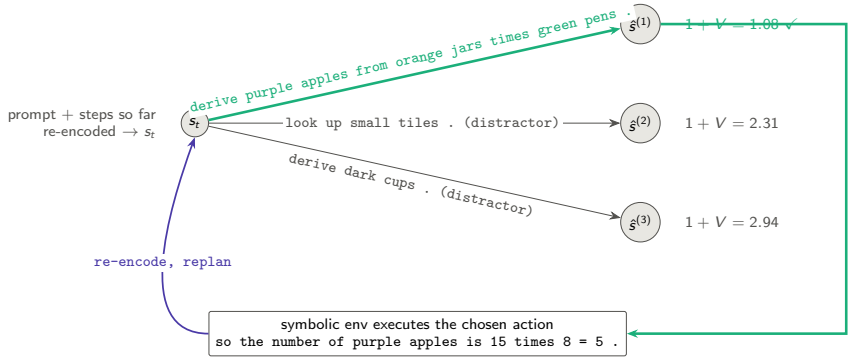
Planning: closed-loop MPC by exhaustive latent search

Interface knowledge (legitimate): feasible actions = unresolved variables whose stated dependencies are resolved — readable from the prompt. **Consequence knowledge**: latent only.

1. Encode prompt + steps so far $\rightarrow s$ (and s_0 once).
2. Enumerate feasible action *sequences* to depth L (lookahead; cap 64), encode their intent phrases $\rightarrow a_i$.
3. Roll out $\hat{s} \leftarrow F(\hat{s}, a_i)$ along each sequence.
4. Score: cost = steps taken + $V(\hat{s}_{\text{end}}, s_0)$ (estimated total steps).
5. Execute the first action of the argmin in the symbolic environment $\rightarrow$ real outcome sentence appended; go to 1.

Success = query resolved within the **strict optimal budget** $|\mathcal{A}_q|$ (slack 0: any distractor pick is fatal). 200 val episodes. Discrete enumeration replaces CEM; a sampled action prior would need CEM/MPPI.

Planning: search graph (one replanning round, lookahead 1)



V values are predicted *remaining necessary steps* of the predicted next latent; lookahead 2 scores 2-step latent rollouts.

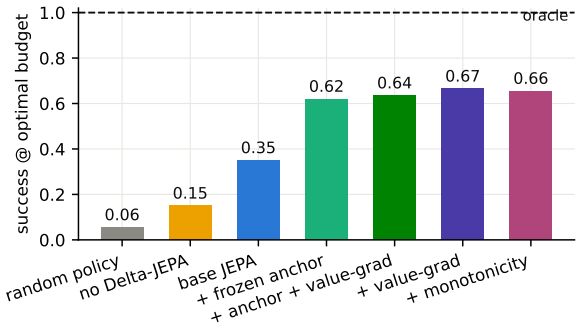
Experiment grid (each = one training run, 20 epochs)

run	$\mathcal{L}_{\text{chunk}}$ anchor	value grad → encoder	other
`disc_base`	—	no	reference
`disc_no_delta`	—	no	$\mathcal{L}_{\text{act}}$ weight 0
`disc_chunkpred`	frozen, w=2.0	no	—
`disc_valgrad`	—	**yes**	—
`disc_combo`	frozen, w=2.0	**yes**	—
`disc_value_only`	—	**yes**	*all* JEPA weights 0
`disc_frozen_enc`	—	no	encoders frozen at init
`disc_straight_{lo,mid,---}`	frozen, w=2.0	**yes**	$+\,\mathcal{L}_{\text{curv}}$, $\lambda \in \{.02, .05, .1\}$
`disc_mono` / `_hi`	frozen, w=2.0	**yes**	$+\,\mathcal{L}_{\text{mono}}$, w1 m.02 / w2 m.05
`disc_geo_lo` / `_geo`	frozen, w=2.0	**yes**	both regularizers
`disc_mono_novalue`	frozen, w=2.0	no	$+\,\mathcal{L}_{\text{mono}}$, *no value head*

Baselines at plan time: **random feasible policy**, **symbolic oracle** (=100%), and **oracle goal-distance energy** (LeWM-style, uses the encoded solved state — diagnostic only).

Result 1 — training objectives determine plannability

Claim: JEPA losses turn a frozen-vocabulary encoder into a plannable world model; each added mechanism helps.



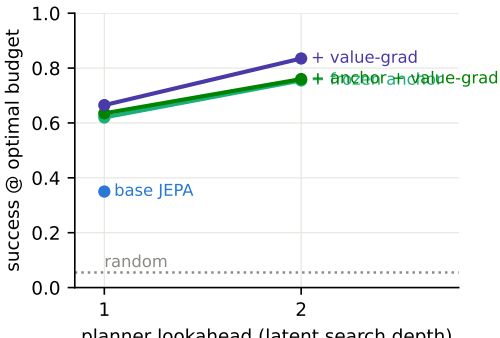
Success within the *strict optimal budget*, greedy lookahead 1, 200 episodes, mean necessary steps 4.07.

With slack 2: base 0.75, +anchor 0.88, +value-grad 0.92, +monotonicity **0.935** (random 0.41).

Distractor pick rate: random 0.43 → base 0.28 → valgrad 0.14.

Result 2 — deeper latent search helps monotonically

Claim: Success grows with search depth: rollouts of F and the energy stay trustworthy over multiple imagined steps.



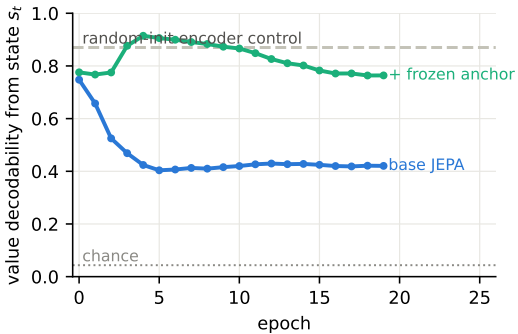
Lookahead 2 = exhaustive 2-step feasible sequences (cap 64), scored at the imagined end state.

`disc_valgrad`: 0.665 → **0.835**; distractor rate 0.14 → 0.07.

This is the planning-time compute knob promised by latent world models — no retraining involved.

Result 3 — encoder–predictor collusion, measured

Claim: Pure state-prediction JEPA *removes* outcome information that was linearly present at initialization; a frozen outcome anchor prevents it.



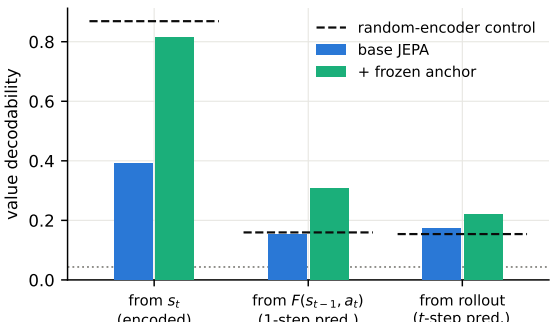
Probe: linear decodability of the just-computed value (23 classes) from s_t, val split.

Both loss sides are learned; EMA targets *trail the online encoder*, so deleting a hard-to-predict feature lowers the loss — stop-grad does not pin content.

Consequence in `disc_base`: answer decodable from the terminal state at only 0.26 (random-init control: 0.65).

Result 4 — latent arithmetic under the frozen anchor

Claim: With $\mathcal{L}_{\text{chunk}}$, the predictor computes outcomes it never observes: value decodability from *predicted* latents doubles the random-encoder control.



Dashed = random-init encoder control (surface memory, no computation); dotted = chance $1/23$.

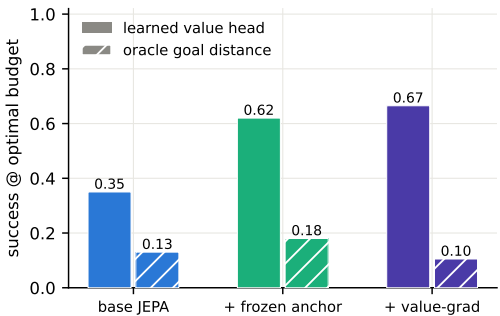
From $F(s_{t-1}, a_t)$: base 0.15 (= control) → anchor **0.31**.

Answer from terminal state: 0.26 → **0.98**.

Full-trace open-loop rollout: 0.22 (error compounds; matches Result 2's need for re-encoding).

Result 5 — the goal energy must be learned (LeWM test)

Claim: Raw latent distance to the oracle goal state plans barely above random; the trained value head is the energy that works.



Oracle goal = encoded terminal state of the *true* solution (planner gets it for free — upper-bound diagnostic).

Scoring $\|\mathrm{LN}(\hat{s}) - \mathrm{LN}(s_{\text{goal}})\|_1$: base 0.13, anchor 0.18, valgrad 0.105 (distractor rates $\approx$ random).

$\Rightarrow$ *unregularized* JEPA geometry is not a distance-to-goal metric. Result 7 shows temporal straightening fixes exactly this.

Result 6 — Delta-JEPA ablation; stability

Claim: Displacement-level action decoding is load-bearing for planning (replicates Delta-JEPA, arXiv:2606.31232, in language).

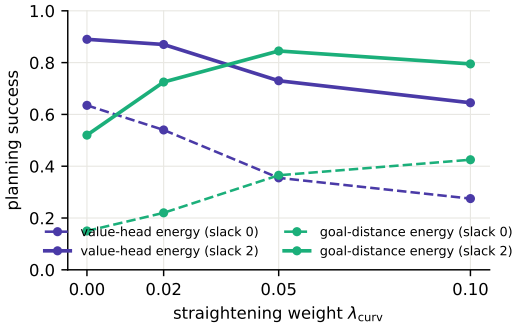
	success@opt	success@slack2	distractor rate	op from Δs
`disc_base`	**0.35**	**0.75**	0.28	1.00
`disc_no_delta`	0.15	0.51	0.40	0.98*

*op stays decodable (it correlates with surface features) but transitions lose the *action-discriminative* structure the planner needs.

Stability (all runs): per-dim state std ≈ 1.03, RankMe effective rank 220–243 / 256, no collapse; VICReg + EMA sufficed, discrete codes (FSQ) not needed.

Result 7 — temporal straightening makes raw geometry plannable

Claim: The curvature loss (arXiv:2603.12231) turns latent distance into a progress metric: goal-distance planning goes 0.15 → 0.43 @optimal (0.52 → 0.85 @slack 2) — but it *trades off* against the value head.



Dose-response over $\lambda_{\text{curv}} \in \{0, .02, .05, .1\}$ (all on the combo recipe):

	0	.02	.05	.1
velocity cos	−.25	−.01	.42	.67
corr(d^{goal}, remaining)	.73	.75	.80	.86
value probe from s_t	.89	.89	.74	.43

Geometry improves exactly as content decodability degrades; $\lambda = .05$ is the raw-geometry sweet spot.

Result 8 — two energies, one metric: the content–geometry trade-off

Claim: Straightening buys geometry planning, the monotonicity hinge keeps content and sets the value-planning record; geometry needs JEPA losses, not value labels.

run	goal-distance energy @opt	goal-distance energy @slack2	value-head energy @opt	value-head energy @slack2
`disc_combo` ($\lambda = 0$)	0.15	0.52	0.635	0.89
`disc_straight_mid` (λ=.05)	0.365	**0.845**	0.355	0.73
`disc_mono_hi` (hinge w2 m.05)	0.255	0.68	**0.655**	**0.935**
`disc_mono_novalue` (*no* V)	0.20	0.65	—	—
`disc_value_only` (*no* JEPA)	0.06	—	0.075	0.43
random	0.06	0.095	0.06	0.095

Three readings. (i) One latent metric cannot currently maximize both energies — a learned *projection head* for the geometric energy is the obvious fix. (ii) A model with **no value head** still plans at 0.65 @slack 2 from pure geometry. (iii) `disc_value_only`'s geometry is *exactly random*: planning-relevant structure comes from the JEPA objectives, not the value labels.

Probe suite (champion vs. reference; random-encoder control)

probe (linear, val split)	`base`	`chunkpred`	random-enc
value of current step from s_t	0.39	**0.81**	0.87
value from $F(s_{t-1}, a_t)$ (latent arithmetic)	0.15	**0.31**	0.15
value from open-loop rollout	0.17	**0.22**	0.15
op type from Δs	1.00	1.00	0.97
step goal-relevance from Δs	0.89	**0.93**	0.89
remaining steps from s_t	0.44	**0.61**	0.41
resolved count from s_t	0.97	0.92	0.95
final answer from terminal state	0.26	**0.98**	0.62
answer from prompt alone s_0 (should be hard)	0.035	0.05	0.04
value-head MAE (remaining steps)	0.76	0.57	—

`disc_combo` adds value-grad: MAE **0.38** with the same collusion-free probes (state value 0.89).

Honest limitations & what this is not

- **iGSM-style, not iGSM**: flat DAG, binary mod-23 ops, one question template, templated English — no hierarchical categories / instance-vs-abstract parameters of the original benchmark.
- **Actions are observed intents**, not variational latents: the symbolic world offers an enumerable interface. The variational prior $p(a \mid s, g)$ (for CEM/MPPI over sampled actions) is future work.
- **The best goal energy is still supervised** (remaining-steps labels); straightened raw geometry closes much of the gap (0.845 vs 0.935 @slack 2, Results 7–8) and the curvature loss is *label-free*, but the monotonicity hinge uses the same labels as the value head.
- Value labels shape only the value head unless `valgrad`; the op-CE in LDAD uses the executed action's type (JEPA-legit: actions are observed at training time).
- Text "generation" is action selection + template rendering; a learned renderer is orthogonal to the world model.

Next steps

- **Decouple the two energies**: straighten a learned *projection* $\pi(s)$ instead of s itself, so goal-distance planning and content decodability stop competing (Result 8).
- **Advantage head** $A(s, a)$ trained on Δ(remaining) and action *ranking* losses — sharper than $V(F(s, a))$, which conflates predictor and value error.
- **Hierarchical planning**: use the trained F_{hi} + macro codes to propose 3-step jumps, refine with F (HWM).
- **Unobserved actions**: variational $q(a \mid s_{t-1}, s_t)$ with KL to prior $p(a \mid s, g)$; inspect emergent action clusters.
- **Harder worlds**: faithful iGSM (hierarchy + category sums), larger moduli, paraphrase templates, natural-language noise; scale.